Chiral relations for entanglement in quantum self phase modulation

M. Gil de Oliveira* and A. Z. Khoury
Instituto de Física, Universidade Federal Fluminense, 24210-346 Niterói, RJ, Brazil
(Dated: July 13, 2026)

An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. SPATIALLY STRUCTURED BOSONIC FIELD

Consider a bosonic quantum field $\mathcal{E}(\mathbf{r})$ satisfying the commutation relations

$$\left[\mathcal{E}(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)\right] = \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2),\ \left[\mathcal{E}(\mathbf{r}_1), \mathcal{E}(\mathbf{r}_2)\right] = \left[\mathcal{E}^\dagger(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)\right] = 0 \tag{1}$$

In the above equation and in what follows, $\mathbf{r} = (x, y)$ is the transverse coordinate.

One can introduce ladder operators $a_v, a_v^\dagger$ associated with the spatial mode v by the expression

$$a(v) = \int_{\mathbb{R}^2} v^*(\mathbf{r})\mathcal{E}(\mathbf{r})d\mathbf{r}. \tag{2}$$

We then have

$$\left[a(v_1), a(v_2)^\dagger\right] = \int_{\mathbb{R}^2} v_1^*(\mathbf{r})v_2(\mathbf{r})d\mathbf{r} = \langle v_1|v_2\rangle\,, \tag{3}$$

while $[a(v_1), a(v_2)] = \left[a(v_1)^\dagger, a(v_2)^\dagger\right] = 0$.

Let us define the quadrature associated to the spatial mode v as

$$\begin{aligned} X(v) &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}^2} v^*(\mathbf{r})\mathcal{E}(\mathbf{r}) + v(\mathbf{r})\mathcal{E}^\dagger(\mathbf{r})d\mathbf{r} \\ &= \frac{a(v) + a(v)^\dagger}{\sqrt{2}} \end{aligned} \tag{4}$$

We then have the commutation relation

$$[X(v_1), X(v_2)] = i\,\mathrm{Im}\,\langle v_1|v_2\rangle\,, \tag{5}$$

which leads to the uncertainty principle

$$\Delta X(v_1)^2 \Delta X(v_2)^2 \geq \frac{|\mathrm{Im}\,\langle v_1|v_2\rangle|^2}{4}. \tag{6}$$

One could also introduce the complementary quadrature $P(v) = X(-iv)$ so that

$$[X_{v_1}, P_{v_2}] = i\,\mathrm{Re}\,\langle v_1|v_2\rangle \tag{7}$$

and

$$\Delta X(v_1)^2 \Delta P(v_2)^2 \geq \frac{|\mathrm{Re}\,\langle v_1|v_2\rangle|^2}{4}. \tag{8}$$

We are interested in observables of the form

$$\Gamma(v_1, v_2) = \frac{1}{2}\left\langle \{X(v_1), X(v_2)\}\right\rangle - \langle X(v_1)\rangle \langle X(v_2)\rangle = \mathrm{Re}\left[\frac{\langle v_1|v_2\rangle}{2} + f(v_1, v_2) + h(v_1, v_2)\right], \tag{9}$$

* marcosgil@id.uff.br

where we have defined

$$\begin{aligned} f(v_1,v_2) &= \langle a(v_1)a(v_2)\rangle - \langle a(v_1)\rangle \langle a(v_2)\rangle \\ h(v_1,v_2) &= \left\langle a(v_2)^\dagger a(v_1)\right\rangle - \left\langle a(v_2)^\dagger\right\rangle \langle a(v_1)\rangle \end{aligned}. \tag{10}$$

In terms of the fields, the required quantities are written as

$$f(v_1,v_2) = \int v_1^*(\mathbf{r}_2)v_2^*(\mathbf{r}_1)F(\mathbf{r}_1,\mathbf{r}_2)d\mathbf{r}_1 d\mathbf{r}_2; \quad F(\mathbf{r}_1,\mathbf{r}_2) = \langle \mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\rangle - \langle \mathcal{E}(\mathbf{r}_1)\rangle \langle \mathcal{E}(\mathbf{r}_2)\rangle \tag{11}$$

$$h(v_1,v_2) = \int v_1^*(\mathbf{r}_1)v_2(\mathbf{r}_2)H(\mathbf{r}_1,\mathbf{r}_2)d\mathbf{r}_1 d\mathbf{r}_2; \quad H(\mathbf{r}_1,\mathbf{r}_2) = \left\langle \mathcal{E}^\dagger(\mathbf{r}_2,z)\mathcal{E}(\mathbf{r}_1,z)\right\rangle - \left\langle \mathcal{E}^\dagger(\mathbf{r}_2,z)\right\rangle \langle \mathcal{E}(\mathbf{r}_1,z)\rangle \tag{12}$$

This formulation is useful because f and g are normally ordered expectation values, which can be directly obtained from stochastic simulations in the positive P representation. Furthermore, this representation also makes it easier to perform linear combinations of the modes, which will be useful later. These combinations will be based on the properties

$$\begin{aligned} f(v_1,v_2+\lambda v_3) &= f(v_1,v_2) + \lambda^* f(v_1,v_3) \\ f(v_1,v_2) &= f(v_2,v_1) \\ h(v_1,v_2+\lambda v_3) &= h(v_1,v_2) + \lambda h(v_1,v_3) \\ h(v_1,v_2) &= h(v_2,v_1)^* \end{aligned} \tag{13}$$

which are a reflection of the fact that $a(v)$ is antilinear while $a^\dagger(v)$ is linear.

A. One mode correlation

Let us analyze the correlation matrix $\gamma_{jk} = \frac{1}{2}\langle\{Z_j,Z_k\}\rangle - \langle Z_j\rangle\langle Z_k\rangle$ where $\mathbf{Z} = (X(v),P(v))$. We observe that $h(v,v)\in\mathbb{R}$, so we get

$$\gamma = \begin{pmatrix} \frac{1}{2}+h(v,v) & 0 \\ 0 & \frac{1}{2}+h(v,v) \end{pmatrix} + \begin{pmatrix} \operatorname{Re} f(v,v) & -\operatorname{Im} f(v,v) \\ -\operatorname{Im} f(v,v) & -\operatorname{Re} f(v,v) \end{pmatrix}. \tag{14}$$

Notice that $f(e^{i\phi}v,e^{i\phi}v) = e^{-i2\phi}f(v,v)$. Therefore, we can diagonalize γ by choosing $\phi = [\arg f(v,v)\pm\pi]/2$. We then get eigenvalues

$$\lambda_\pm = \frac{1}{2} + h(v,v) \pm |f(v,v)|. \tag{15}$$

We then see that the quadrature rotated along the axis ϕ has minimum variance of λ_-, while the one rotated along the axis $\phi+\pi/2$ has maximum variance λ_+. Furthermore, we have squeezing if $|f(v,v)| > h(v,v)$.

B. Duan Criterion

Consider two orthonormal states v_1, v_2, i.e., $\langle v_j|v_k\rangle = \delta_{jk}$. The Duan criterion [1] states that if

$$D = \frac{1}{2}\left\langle[\Delta X(v_1)+\Delta X(v_2)]^2\right\rangle + \frac{1}{2}\left\langle[\Delta P(v_1)-\Delta P(v_2)]^2\right\rangle < 1, \tag{16}$$

then the states are entangled.

Let us define $v_\pm = (v_1\pm v_2)/\sqrt{2}$. We then have

$$D = \left\langle \Delta X(v_+)^2\right\rangle + \left\langle \Delta P(v_-)^2\right\rangle. \tag{17}$$

Applying (9) we obtain

$$\begin{aligned} D &= 1 + h(v_+,v_+) + h(v_-,v_-) + \operatorname{Re}\left[f(v_+,v_+) - f(v_-,v_-)\right] \\ &= 1 + h(v_1,v_1) + h(v_2,v_2) + 2\operatorname{Re} f(v_1,v_2) \end{aligned} \tag{18}$$

Once again, we can maximize the possible violation by rotating the quadrature of the modes, say, v_1 by an angle $\phi = \left[\arg f(v_1,v_2)\pm\pi\right]/2$. We get an optimal possible violation of

$$D_{\text{opt}} = 1 + h(v_1,v_1) + h(v_2,v_2) - 2|f(v_1,v_2)| \tag{19}$$

II. THE KERR SYSTEM

A. Heisenberg Equation for the Field and Linear Solution

We will assume that the field is propagating along the z direction, and satisfies the Heisenberg equation of motion

$$2ik\partial_z\mathcal{E}(\mathbf{r},z)+\nabla^2\mathcal{E}(\mathbf{r},z)=g\mathcal{E}^\dagger(\mathbf{r},z)\mathcal{E}^2(\mathbf{r},z). \tag{20}$$

Let us decompose $\mathcal{E}=\mathcal{E}_0+\delta\mathcal{E}$ where $\mathcal{E}_0$ is the solution of

$$2ik\partial_z\mathcal{E}_0(\mathbf{r},z)+\nabla^2\mathcal{E}_0(\mathbf{r},z)=0,\quad \mathcal{E}_0(\mathbf{r},0)=\mathcal{E}(\mathbf{r},0). \tag{21}$$

Let u be a solution of the paraxial wave equation $2ik\partial_z u(\mathbf{r},z)+\nabla^2 u(\mathbf{r},z)=0$. Then, the anihilation operator corresponding to mode u is given by

$$a(u,z)=\int u^*(\mathbf{r},z)\mathcal{E}(\mathbf{r},z)d\mathbf{r}=a_0(u)+\delta a(u,z) \tag{22}$$

where

$$a_0(u)=\int u^*(\mathbf{r},z)\mathcal{E}_0(\mathbf{r},z)d\mathbf{r} \tag{23}$$

is independent of z (take a z derivative, use de equations of motion and integrate by parts) and

$$\delta a(u,z)=\int u^*(\mathbf{r},z)\delta\mathcal{E}(\mathbf{r},z)d\mathbf{r} \tag{24}$$

Our initial state will be a coherent one:

$$|\alpha\rangle=D(\alpha)|0\rangle=\exp\left(a(\alpha)^\dagger-a(\alpha)\right)|0\rangle\,. \tag{25}$$

Here, $\alpha(\mathbf{r})$ is a non normalized spatial function, with norm $\|\alpha\|$. This state is an eigenvector of the free field:

$$\mathcal{E}_0(\mathbf{r},z)|\alpha\rangle=\alpha(\mathbf{r},z)|\alpha,u\rangle \tag{26}$$

The variation $\delta\mathcal{E}$ satisfies

$$2ik\partial_z\delta\mathcal{E}(\mathbf{r},z)+\nabla^2\delta\mathcal{E}(\mathbf{r},z)=g\mathcal{E}^\dagger(\mathbf{r},z)\mathcal{E}^2(\mathbf{r},z). \tag{27}$$

Evaluating this at $z=0$ gives us

$$2ik\partial_z\delta\mathcal{E}(\mathbf{r},0)=g\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \tag{28}$$

from which we obtain the Taylor expansion

$$\delta\mathcal{E}(\mathbf{r},z)\approx-\frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \tag{29}$$

For the complete field we then have

$$\mathcal{E}(\mathbf{r},z)\approx\mathcal{E}_0(\mathbf{r},0)-\frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \tag{30}$$

This expression will be the basis of our further developments.

B. Expectation value of field

Taking expectation values, we arrive at

$$\langle\mathcal{E}(\mathbf{r},z)\rangle\approx\alpha(\mathbf{r},0)-\frac{igz}{2k}|\alpha(\mathbf{r},0)|^2\alpha(\mathbf{r},0). \tag{31}$$

The expectation value of the annihilation operator a_v corresponding to a spatial mode v is then

$$\langle a_v \rangle = \langle v|\alpha\rangle - \frac{igz}{2k} \langle v, \alpha|\alpha, \alpha\rangle \tag{32}$$

In the above expression we have introduced the second spatial overlap

$$\langle v_1, v_2|u_1, u_2\rangle = \int v_1^*(\mathbf{r}, z) v_2^*(\mathbf{r}, z) u_1(\mathbf{r}, z) u_2(\mathbf{r}, z) d\mathbf{r}. \tag{33}$$

When $\alpha = \|\alpha\| u_{0l}$ is a Laguerre-Gaussian mode with radial order $p = 0$ and topological charge l, we have that

$$|u_{0l}(\mathbf{r}, z)| u_{0l}^2(\mathbf{r}, z) = \sum_{q=0}^{|l|} C_{ql} \overline{u}_{ql}(\mathbf{r}, z) \tag{34}$$

where $\overline{u}_{ql}$ are Laguerre-Gaussian modes with a reduced waist. Therefore, one may write

$$\langle a_v \rangle = \|\alpha\| \langle v|u_{0,l}\rangle - \frac{igz\|\alpha\|^3}{2k} \sum_{q=0}^{|l|} C_{ql} \langle v|\overline{u}_{ql}\rangle \tag{35}$$

In particular, when $v = \overline{u}_{ql}$, we have that

$$\langle a_{\overline{u}_{ql}} \rangle = \begin{cases} \|\alpha\| \langle \overline{u}_{ql}|u_{0,l}\rangle - \dfrac{igz\|\alpha\|^3 C_{ql}}{2k} & q \leq |l| \\ \|\alpha\| \langle \overline{u}_{ql}|u_{0,l}\rangle & q > |l| \end{cases}. \tag{36}$$

C. Correlation functions

One of the two point field function is

$$\mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z)\delta\mathcal{E}(\mathbf{r}_2, z) + \delta\mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) \tag{37}$$

where we have only kept terms linear in $\delta\mathcal{E}$.

Close to $z = 0$, we have that

$$\mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1, 0)\mathcal{E}_0(\mathbf{r}_2, 0) - \frac{igz}{2k}\left[\mathcal{E}_0^\dagger(\mathbf{r}_1, 0)\mathcal{E}_0^\dagger(\mathbf{r}_2, 0)\mathcal{E}_0^2(\mathbf{r}_2, 0) - \mathcal{E}_0^\dagger(\mathbf{r}_1, 0)^2\mathcal{E}_0(\mathbf{r}_1, 0)\mathcal{E}_0(\mathbf{r}_2, 0)\right] \tag{38}$$

Evaluating this at our initial state gives us

$$\begin{aligned} \langle \mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z)\rangle &\approx \alpha^*(\mathbf{r}_1, 0)\alpha(\mathbf{r}_2, 0)\left\{1 - \frac{igz}{2k}\left[|\alpha(\mathbf{r}_2, 0)|^2 - |\alpha(\mathbf{r}_1, 0)|^2\right]\right\} \\ &\approx \langle \mathcal{E}^\dagger(\mathbf{r}_1, z)\rangle \langle \mathcal{E}(\mathbf{r}_2, z)\rangle \end{aligned} \tag{39}$$

where the last approximate inequality comes from direct comparison with (31). In particular, this shows that, for any v_1 and v_2, we have $H(\mathbf{r}_1, \mathbf{r}_2) \approx 0$.

We can also use this to write the expectation value of the number operator:

$$\langle n(v)\rangle = \langle a^\dagger(v)a(v)\rangle \approx |\langle a(v)\rangle|^2 \approx |\langle v|\alpha\rangle|^2 + \frac{gz}{k} \operatorname{Im} \langle v|\alpha\rangle \langle \alpha, \alpha|\alpha, v\rangle \tag{40}$$

When both α and v are arbitrary Laguerre-Gaussian modes (any p, l, w), we have that $\operatorname{Im} \langle \alpha|u\rangle \langle \alpha, \alpha|\alpha, v\rangle = 0$, so there is no first order correction to the occupation number.

The other two point field function is

$$\begin{aligned} \mathcal{E}(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z) &\approx \mathcal{E}_0(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) + \mathcal{E}_0(\mathbf{r}_1, z)\delta\mathcal{E}(\mathbf{r}_2, z) + \delta\mathcal{E}(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) \\ &\approx \mathcal{E}_0(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) - \frac{igz}{2k}\left[\mathcal{E}_0(\mathbf{r}_1, z)\mathcal{E}_0^\dagger(\mathbf{r}_2, z)\mathcal{E}_0^2(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z)\mathcal{E}_0^2(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z)\right] \\ &= \mathcal{E}_0(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) - \frac{igz}{2k}\left[\mathcal{E}_0^\dagger(\mathbf{r}_2, z)\mathcal{E}_0(\mathbf{r}_1, z)\mathcal{E}_0^2(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z)\mathcal{E}_0^2(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) + \delta(\mathbf{r}_1 - \mathbf{r}_2)\mathcal{E}_0^2(\mathbf{r}_2, z)\right] \end{aligned} \tag{41}$$

The appearance of the delta function here is important, because now

$$\begin{aligned}\langle \mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\rangle &\approx \alpha(\mathbf{r}_1,z)\alpha(\mathbf{r}_2,z)\left[1-\frac{igz}{2k}\left[|\alpha(\mathbf{r}_2,z)|^2+|\alpha(\mathbf{r}_1,z)|^2\right]\right]-\frac{igz}{2k}\delta(\mathbf{r}_1-\mathbf{r}_2)\alpha^2(\mathbf{r}_2,z)\\ &\approx \langle \mathcal{E}(\mathbf{r}_1,z)\rangle\langle \mathcal{E}(\mathbf{r}_2,z)\rangle-\frac{igz}{2k}\delta(\mathbf{r}_1-\mathbf{r}_2)\alpha^2(\mathbf{r}_2,z)\end{aligned} \tag{42}$$

Therefore, we may conclude that

$$F(\mathbf{r}_1,\mathbf{r}_2)\approx -\frac{igz}{2k}\delta(\mathbf{r}_1-\mathbf{r}_2)\alpha^2(\mathbf{r}_2,z), \tag{43}$$

$$f(v_1,v_2)\approx -\frac{igz}{2k}\langle v_1,v_2|\alpha,\alpha\rangle, \tag{44}$$

and

$$\Gamma(v_1,v_2)=\frac{\mathrm{Re}\,\langle v_1|v_2\rangle}{2}+\frac{gz}{2k}\,\mathrm{Im}\,\langle v_1,v_2|\alpha,\alpha\rangle \tag{45}$$

This leads us to a minimum quadrature variance for mode v of

$$\lambda_-=\frac{1}{2}-\frac{z}{2k}|g\langle v,v|\alpha,\alpha\rangle| \tag{46}$$

and an optimal Duan parameter for modes v_1 and v_2 of

$$D_{\mathrm{opt}}=1-\frac{z}{k}|g\langle v_1,v_2|\alpha,\alpha\rangle|. \tag{47}$$

The corresponding angles are $\phi_{\mathrm{squeezing}}=(-\pi/2\pm\pi+\arg g\langle v,v|\alpha,\alpha\rangle)/2$ and $\phi_{\mathrm{duan}}=(-\pi/2\pm\pi+\arg g\langle v_1,v_2|\alpha,\alpha\rangle)/2$

At this point, it becomes clear that the overlaps of the form $\langle v_1,v_2|\alpha,\alpha\rangle$ are the ones that dominate this sort of process.

Consider the case where all the involved modes are vortices: $u(\mathbf{r})=u_r(r)e^{il_u\phi}$ and similarly for the v functions. We have squeezing if $l_v=l_u$ and we have entanglement if $l_{v_1}+l_{v_2}=2l_u$. If all the radial functions are positive, as is the case for $LG_{0,0}$ and $LG_{0,\pm1}$, and α is positive, then the optimal angles are both $\mathrm{sign}(g)\pi/4$.

D. Selection Rules for two-mode entanglement in the first order treatment

Consider the case where the input mode is $u(\mathbf{r})=LG_{0,l}(\mathbf{r})$, where

$$LG_{pl}(\mathbf{r};w)=(-1)^p\frac{\mathcal{N}_{pl}}{w}\left(\frac{2r^2}{w^2}\right)^{|l|/2}L_p^{|l|}\left(\frac{2r^2}{w^2}\right)e^{-r^2/w^2+il\phi} \tag{48}$$

is a Laguerre-Gaussian mode at the waist plane with radial order p, topological charge l and waist w. The normalization constant is

$$\mathcal{N}_{pl}=\sqrt{\frac{2}{\pi}}\sqrt{\frac{p!}{(p+|l|)!}} \tag{49}$$

Furthermore, let $v_2\propto LG_{0,l_2}(\mathbf{r};w_2)$ where w_2 is free and l_2 is only constrained by $l_2\neq l$. Now, define $l_1=2l-l_2$ and $1/w_1^2=2/w^2+1/w_2^2$. We then have that

$$\begin{aligned}v_2^*(\mathbf{r})u^2(\mathbf{r}) &\propto \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l}^2}{w_2w^2}\left(\frac{2r^2}{w_2^2}\right)^{|l_2|/2}\left(\frac{2r^2}{w^2}\right)^{|l|}e^{-r^2/w_1^2+il_1\phi}\\ &=\frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l}^2w_1^{2|l|+|l_2|}}{w_2^{|l_2|+1}w^{2|l|+2}}\left(\frac{2r^2}{w_1^2}\right)^{|l_1|/2}\left(\frac{2r^2}{w_1^2}\right)^{p_{\max}}e^{-r^2/w_1^2+il_1\phi}\end{aligned} \tag{50}$$

where we have defined

$$p_{\max} = \frac{1}{2}(2|l| + |l_2| - |2l - l_2|) = \begin{cases} 0, & ll_2 \le 0 \\ \min(2|l|, |l_2|), & ll_2 > 0 \end{cases} \tag{51}$$

Then, by expanding the monomial that is raised to the power of $p_{\max}$ in terms of Laguerre polynomials, we get

$$v_2^*(\mathbf{r})u^2(\mathbf{r}) = \frac{1}{w^2}\sum_{p=0}^{p_{\max}} C_p LG_{pl_1}(\mathbf{r}; w_1) \tag{52}$$

where

$$C_p = \frac{p_{\max}!\mathcal{N}_{0l_2}\mathcal{N}_{0l}^2 w_1^{2|l|+|l_2|+1}}{\mathcal{N}_{pl_1} w_2^{|l_2|+1} w^{2|l|}} \binom{p_{\max} + |l_1|}{p_{\max} - p} \tag{53}$$

Therefore, setting $v_1 = e^{i\phi}LG_{pl_1}(\mathbf{r}; w_1)$, which is orthogonal to v_2, and considering the two mode subsystem formed by v_1, v_2, we get an optimal Duan violation of

$$D_{\text{opt}} = \begin{cases} 1 - \dfrac{z\|\alpha\|^2 C_p}{kw^2}, & p \le p_{\max} \\ 1, & p > p_{\max} \end{cases} \tag{54}$$

Therefore, when $p \le p_{\max}$, we have bipartite entanglement in this two-mode subsystem.

E. Multimode entanglement

III. NUMERICAL CONSIDERATIONS

Note that, according to (1), the field $\mathcal{E}(\mathbf{r})$ has dimensions of $1/\sqrt{\text{area}}$. Therefore, by dimensional analysis on eq. (20), we have that g is dimensionless.

In order to put the equation in a more convenient form, we introduce the dimensionless coordinate $\tilde{\mathbf{r}} = \mathbf{r}/w_0$ where w_0 is the waist of the input beam. We also introduce the dimensionless propagation distance $\tilde{z} = z/z_R$ where $z_R = kw_0^2/2$ is the Rayleigh range. Finally, we introduce the rescaled field $\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) = \sqrt{\Delta A}\mathcal{E}(\mathbf{r}, z)$, where ΔA is the area of a pixel in the numerical grid. In terms of these variables, the commutation relations read

$$\left[\tilde{\mathcal{E}}(\tilde{\mathbf{r}}), \tilde{\mathcal{E}}(\tilde{\mathbf{r}})^\dagger\right] \approx \delta_{\tilde{\mathbf{r}},\tilde{\mathbf{r}}'}, \tag{55}$$

while the equation of motion reads

$$i\partial_{\tilde{z}}\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) = -\frac{1}{4}\tilde{\nabla}^2\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) + \frac{g}{4\Delta\tilde{A}}\tilde{\mathcal{E}}^\dagger(\tilde{\mathbf{r}}, \tilde{z})\tilde{\mathcal{E}}^2(\tilde{\mathbf{r}}, \tilde{z}) \tag{56}$$

where $\Delta\tilde{A} = \Delta A/w_0^2$ is the area of a pixel in the rescaled coordinates.

In the positive P representation, the field $\tilde{\mathcal{E}}$ is represented by two independent complex stochastic fields ψ_1 and ψ_2 satisfying the stochastic equations (compare with [2])

$$\begin{aligned} i\partial_{\tilde{z}}\psi_1(\tilde{\mathbf{r}}, \tilde{z}) &= -\frac{1}{4}\tilde{\nabla}^2\psi_1(\tilde{\mathbf{r}}, \tilde{z}) + \frac{g}{4\Delta\tilde{A}}\psi_2(\tilde{\mathbf{r}}, \tilde{z})\psi_1^2(\tilde{\mathbf{r}}, \tilde{z}) + \sqrt{\frac{ig}{4\Delta\tilde{A}}}\psi_1(\tilde{\mathbf{r}}, \tilde{z})\xi_1(\tilde{\mathbf{r}}, \tilde{z}) \\ -i\partial_{\tilde{z}}\psi_2(\tilde{\mathbf{r}}, \tilde{z}) &= -\frac{1}{4}\tilde{\nabla}^2\psi_2(\tilde{\mathbf{r}}, \tilde{z}) + \frac{g}{4\Delta\tilde{A}}\psi_1(\tilde{\mathbf{r}}, \tilde{z})\psi_2^2(\tilde{\mathbf{r}}, \tilde{z}) + \sqrt{-\frac{ig}{4\Delta\tilde{A}}}\psi_2(\tilde{\mathbf{r}}, \tilde{z})\xi_2(\tilde{\mathbf{r}}, \tilde{z}) \end{aligned} \tag{57}$$

where ξ_1 and ξ_2 are independent real Gaussian white noises with zero mean and correlations

$$\langle\xi_j(\tilde{\mathbf{r}}, \tilde{z})\xi_k(\tilde{\mathbf{r}}', \tilde{z}')\rangle = \delta_{jk}\delta_{\tilde{\mathbf{r}},\tilde{\mathbf{r}}'}\delta_{\tilde{z},\tilde{z}'} \tag{58}$$

For a coherent initial state in the mode u, we have that $\psi_1(\tilde{\mathbf{r}}, 0) = \alpha(\tilde{\mathbf{r}}, 0)$ and $\psi_2(\tilde{\mathbf{r}}, 0) = \alpha^*(\tilde{\mathbf{r}}, 0)$.

IV. PURELY NONLINEAR APPROACH

Let us make a modification to our equations of motion where

$$2ik\partial_z\mathcal{E}(\mathbf{r},z) = \int_{\mathbb{R}^2} g(\mathbf{s})\mathcal{E}^\dagger(\mathbf{r}-\mathbf{s},z)\mathcal{E}(\mathbf{r}-\mathbf{s},z)d\mathbf{s}\ \mathcal{E}(\mathbf{r},z). \tag{59}$$

In this form, the equation has a constant of motion $\mathcal{E}^\dagger(\mathbf{r},z)\mathcal{E}(\mathbf{r},z)$, which leads us to the exact solution

$$\begin{aligned}\mathcal{E}(\mathbf{r},z) &= \exp\left[-\frac{iz}{2k}\int_{\mathbb{R}^2} g(\mathbf{s})\mathcal{E}^\dagger(\mathbf{r}-\mathbf{s},0)\mathcal{E}(\mathbf{r}-\mathbf{s},0)d\mathbf{s}\right]\mathcal{E}(\mathbf{r},0)\\ &=: \exp\left[\int_{\mathbb{R}^2}\left(e^{-izg(\mathbf{s})/2k}-1\right)\mathcal{E}^\dagger(\mathbf{r}-\mathbf{s},0)\mathcal{E}(\mathbf{r}-\mathbf{s},0)d\mathbf{s}\right] : \mathcal{E}(\mathbf{r},0)\end{aligned} \tag{60}$$

In the last line, we used an important result known as the normal ordering theorem [3, 4]. This allows us to simply calculate the expectation value for the field in an initial coherent state:

$$\langle\mathcal{E}(\mathbf{r},z)\rangle = \exp\left[\int_{\mathbb{R}^2}\left(e^{-izg(\mathbf{s})/2k}-1\right)|\alpha(\mathbf{r}-\mathbf{s},0)|^2 d\mathbf{s}\right]u(\mathbf{r},0) \tag{61}$$

We see that (31) is then a first order approximation of (61) with $g(\mathbf{s}) = g_0\delta(\mathbf{s})$. Nonetheless, we see that we get singularities in the full expression when using this form of g. Let us then choose g as proportional to an approximation of the delta function. For definiteness, let us define

$$\delta_0(x,a) = \frac{\Theta(a/2-|x|)}{a}, \tag{62}$$

which approximates the delta distribution as $a \to 0$. Here, $\Theta(x)$ is the step function. Let us denote $\delta_0(\mathbf{r},a) = \delta_0(x,a)\delta_0(y,a)$. We then choose $g(\mathbf{s}) = g_0\delta_0(\mathbf{s})$.

We will usually make the approximation that the support of the response function g is much smaller than the characteristic distances of the mode α. We can then make the approximation

$$\langle\mathcal{E}(\mathbf{r},z)\rangle = \exp\left[G(z)|\alpha(\mathbf{r},0)|^2\right]\alpha(\mathbf{r},0) \tag{63}$$

where

$$G(z) = \int_{\mathbb{R}^2}\left(e^{-izg(\mathbf{s})/2k}-1\right)d\mathbf{s} = a^2\left(e^{-izg_0/2ka^2}-1\right) \tag{64}$$

Notice then that, in the general case, this is quite dependent on the value of a. Nonetheless, when $z|g_0|/2ka^2 \ll 1$, we get

$$G(z) \approx -\frac{izg_0}{2k}, \tag{65}$$

which is independent of a.

Following the result of [4] we can calculate

$$\begin{aligned}F(\mathbf{r}_1,\mathbf{r}_2) \approx&\alpha(\mathbf{r}_1,0)\alpha(\mathbf{r}_2,0)\exp\left[G(z)(|\alpha(\mathbf{r}_1)|^2+|\alpha(\mathbf{r}_2)^2|)\right]\\ &\times\left[e^{-ig(\mathbf{r}_1-\mathbf{r}_2)z/2k+|\alpha(\mathbf{r}_1)|^2G_1(\mathbf{r}_1-\mathbf{r}_2)}-1\right]\end{aligned} \tag{66}$$

where

$$\begin{aligned}G_1(\mathbf{r},z) &= \frac{1}{2}\int_{\mathbb{R}^2}(e^{-izg(\mathbf{s}-\mathbf{r})}-1)(e^{-izg(\mathbf{s})}-1)d\mathbf{s}\\ &= \frac{a^4(e^{-ig_0z/2ka^2}-1)^2}{2}\delta_1(\mathbf{r},a)\\ &\approx -\frac{(g_0z/2k)^2}{2}\delta_1(\mathbf{r},a)\end{aligned} \tag{67}$$

where we have defined

$$\delta_1(x,a) = \frac{(a-|x|)\Theta(a-|x|)}{a^2} \tag{68}$$

as another approximation of the delta function.

Notice, then, that, for modes with variation on a scale much larger than a, the second line of (66) is essentially proportional to $\delta(\mathbf{r}_1 - \mathbf{r}_2)$, and we may write

$$F(\mathbf{r}_1,\mathbf{r}_2) \approx \alpha(\mathbf{r}_1,z)^2\overline{F}(\mathbf{r}_1,a)\delta(\mathbf{r}_1-\mathbf{r}_2) \tag{69}$$

where

$$\alpha(\mathbf{r},z) = \alpha(\mathbf{r},0)e^{-ig_0z|\alpha(\mathbf{r},0)|^2/2k} \tag{70}$$

is the evolved field under the classical Kerr interaction and

$$\overline{F}(\mathbf{r},a) = \int_{\mathbb{R}^2}\left\{\exp\left[-i\frac{g_0z}{2k}\delta_0(\mathbf{s},a) - \frac{1}{2}\left(\frac{g_0z}{2k}\right)^2\delta_1(\mathbf{s},a)|\alpha(\mathbf{r})|^2\right]-1\right\}d\mathbf{s} \tag{71}$$

In general, this function presents a strong dependency on a. Nonetheless, if we also assume that $z^2|g_0|^2|\alpha(\mathbf{r})|^2/4k^2a^2 \ll 1$, then we may expand the exponential in a Taylor series and we get

$$\overline{F}(\mathbf{r},a) = -\frac{g_0z}{2k}\left[i+\frac{g_0|\alpha(\mathbf{r})|^2z}{2k}\right], \tag{72}$$

which does not depend on a. Substituting this result, we obtain

$$F(\mathbf{r}_1,\mathbf{r}_2) \approx -\frac{g_0z}{2k}\left[i+\frac{g_0|\alpha(\mathbf{r})|^2z}{2k}\right]\alpha(\mathbf{r}_1,z)^2\delta(\mathbf{r}_1-\mathbf{r}_2) \tag{73}$$

and

$$f(v_1,v_2) = -i\frac{g_0z}{2k}\langle v_1,v_2|\alpha(z),\alpha(z)\rangle - \left(\frac{g_0z}{2k}\right)^2\langle v_1,v_2,\alpha(z)|\alpha(z),\alpha(z),\alpha(z)\rangle \tag{74}$$

Once again, following [4], we calculate

$$H(\mathbf{r}_1,\mathbf{r}_2) = \alpha^*(\mathbf{r}_1,z)\alpha(\mathbf{r}_2,z)(e^{|\alpha(\mathbf{r}_1)|^2G_2(\mathbf{r}_1-\mathbf{r}_2)}-1) \tag{75}$$

where

$$G_2(\mathbf{r}) = a^4\left|e^{ig_0z/2ka^2}-1\right|^2\delta_1(\mathbf{r},a) \approx \left(\frac{g_0z}{2k}\right)^2\delta_1(\mathbf{r},a) \tag{76}$$

Therefore,

$$H(\mathbf{r}_1,\mathbf{r}_2) \approx \left(\frac{g_0z}{2k}\right)^2|\alpha(\mathbf{r}_1,z)|^4\delta(\mathbf{r}_1-\mathbf{r}_2) \tag{77}$$

and

$$h(v_1,v_2) = \left(\frac{g_0z}{2k}\right)^2\langle v_1,\alpha(z),\alpha(z)|v_2,\alpha(z),\alpha(z)\rangle \tag{78}$$

Notice that $h(v,v) \geq 0$, so it reduces both squeezing and the violation of Duan's inequality.

---

[1] L.-M. Duan, G. Giedke, J. I. Cirac, and P. Zoller, Inseparability criterion for continuous variable systems, Physical review letters **84**, 2722 (2000).

[2] P. Deuar and P. D. Drummond, Correlations in a bec collision: First-principles quantum dynamics with 150 000 atoms, Phys. Rev. Lett. **98**, 120402 (2007).

[3] K. J. Blow, R. Loudon, S. J. D. Phoenix, and T. J. Shepherd, Continuum fields in quantum optics, Phys. Rev. A **42**, 4102 (1990).

[4] K. J. Blow, R. Loudon, and S. J. D. Phoenix, Exact solution for quantum self-phase modulation, J. Opt. Soc. Am. B **8**, 1750 (1991).